Camila Kaskey

Buffalo, NY | (716) 524-1101 | kamilyak@buffalo.edu | linkedin.com/in/camila-kaskey-653ba1385

Education

University at Buffalo, The State University of New York Buffalo, NY
Bachelor of Science in Mechanical Engineering *Expected May 2028*

- Relevant Coursework: Statics, Dynamics, Mechanics of Solids, Differential Equations, Thermodynamics, Engineering Computations (MATLAB), Electrical Engineering Concepts

Engineering Projects

ASME: Rover Tank Tread Drivetrain Sept. 2025 – May 2026

- Designed modular rover track links with a standardized 20 mm pitch in Fusion 360, developing friction-fit and snap-fit linkage concepts, interlocking terminals, and track pad geometries for a 3D-printed tracked rover.
- 3D printed PLA+ track link prototypes, assembled track systems, and refined designs based on prototype testing.
- Conducted five-trial performance testing of five tread patterns in a sand testbed, comparing travel distance under constant motor torque and selecting the final W-shaped tread design with the highest average distance (3.88 m).

ASME: Mechanical Systems Testing Sept. 2025 – May 2026

- Conducted 40+ validation tests on bearings, treads, motors, chassis members, brackets, and payload mechanisms
- Evaluated component performance through slippage, vibration, deflection, rotational efficiency, failure-load, thermal, and repeated-cycle testing using custom-built rigs and 6000 RPM motor-driven setups.
- Analyzed test data to support material selection and design validation, writing an 80+ page technical report.

MATLAB Programming Project: Tuning Fork Vibrations Simulator Jul. 2025 – Aug. 2025

- Developed a MATLAB program implementing Euler-Bernoulli beam theory to simulate tuning fork vibrations for steel, brass, and aluminum across notes A0-A7 (27.5-3520 Hz), calculating dimensions for each tuning fork design.
- Implemented Newton-Raphson root finding, finite-difference discretization, boundary conditions, and matrix-based numerical solvers to compute the first 12 eigenvalues, mode shapes, natural frequencies, and beam deflections.
- Automated tuning fork geometry calculations, material property analysis, cost estimation, CSV export, and engineering visualizations, including mode shape plots and time-dependent beam deflection simulations.

Automated Hamburger Flipper Jan. 2025 – May 2025

- Built a hamburger flipper using Arduino, servo motors, DC motors, laser sensors, and 3D-printed components.
- Programmed Arduino control logic for burger detection, probe movement, spatula flipping, and motor sequencing.
- Fabricated, assembled, and tested the spatula assembly, probe mechanism, and threaded-rod drive system.

Functional Bluetooth Subwoofer Speaker Jan. 2025 – May 2025

- Assembled a Bluetooth 2-way speaker with a woofer, tweeter, Bluetooth amplifier, and 3.5 mm audio input.
- Fabricated the wooden enclosure in the machine shop using cutting, drilling, sanding, and assembly techniques.
- Installed and wired the speaker drivers, amplifier, and Bluetooth module, testing wired and wireless audio playback.

Leadership & Involvement

Event Planner and Coordinator, ASME Sept. 2025 – Present
University at Buffalo Chapter

- Developed a sponsor-facing website showcasing chapter projects, competition results, awards, and sponsors.
- Host GBMs and project meetings, manage attendance, coordinate event setup, and support meeting operations.

President of Pledge Class, Theta Tau Jan. 2025 – May 2025
Zeta Gamma Chapter

- Led a pledge class of 13 students, organizing meetings, assigning responsibilities, and tracking project progress.
- Coordinated four engineering projects, managing materials, machine shop access, presentations, and deadlines.

Skills

Technical: SolidWorks, Fusion 360, MATLAB, Python, Arduino, 3D Printing
Core Competencies: Communication, Critical Thinking, Teamwork, Leadership
Languages: Russian (Fluent), Kazakh (Fluent), English (Fluent), Turkish (Conversational)

Interests

Robotics, Mechanical Engineering, 3D Printing, Product Design, Scientific Research, Travel